Additional #1

Given:

- Additional Problem #2 from HW #4
- vertical elastic deflection below 2% of its length

a) Find: location of max elastic deflection in handles

$d_1 = .02$ m

$d_2 = .2963$ m

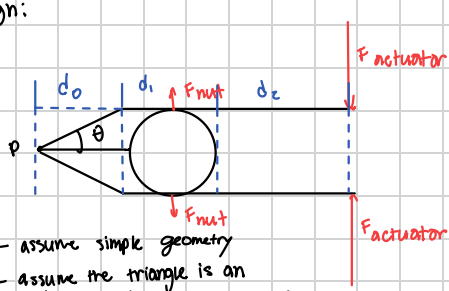
$F_{actuator} = 173.5$ N

$F_{nut} = 2570.5$ N

so, our beam length = .3336 m

and our deflection limit = .3336 m · (.02) = .006672 m

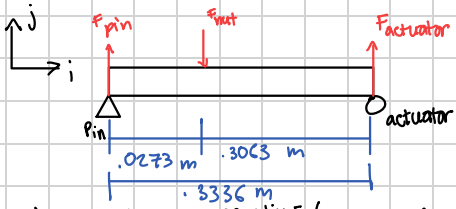
Design:



- assume simple geometry
- assume the triangle is an equilaterial triangle with $L = .02$ m (from the diameter of the nut)
- "height" of triangle is now $\frac{\sqrt{3}}{2} L = .0173$ m (we'll call it d_0)
- assume instance as nut is cracking
- assume actuator acts as a roller because it prevents the handle from moving vertically at the instance of cracking

Beam Design (bottom handle)

- pin @ $x = 0$, F_{nut} @ .0273 m, roller @ .3336 m



Using beam #5 from appendix E (reworked when $a < b$)

$$x_m = \sqrt{\frac{L^2 - a^2}{3}}$$

$$= \sqrt{\frac{(.3336)^2 - (.0273)^2}{3}}$$

$$= .192 \text{ m}$$

Reflection: This problem required us to revaluate our old design to consider our macadamia handles as beams instead of rigid.

b) Find: choose a beam design (cross-section, material) such that the verti elastic deflection is 2% of its length (below)

* assume polycarbonate

Plan:

$$y_{max} = \frac{-Pa\,(L^2-a^2)^{3/2}}{9\sqrt{3}\,EIL}$$

$$-.006672 = \frac{-(2570.5)(.0273)\left[(.3336)^2-(.0273)^2\right]^{3/2}}{9\sqrt{3}\cdot 2.4\times10^9\,Pa\cdot(.3336)\cdot I}$$

$$-.006672 = \frac{-2.579}{1.24807^{10}\,I}\cdot(2 \quad "I)$$

$$-8327151347\,I = -2.579$$

$$I = 3.097\times10^{-8}\,m^4$$

$$= .03097\,(10^6\,mm^4)$$

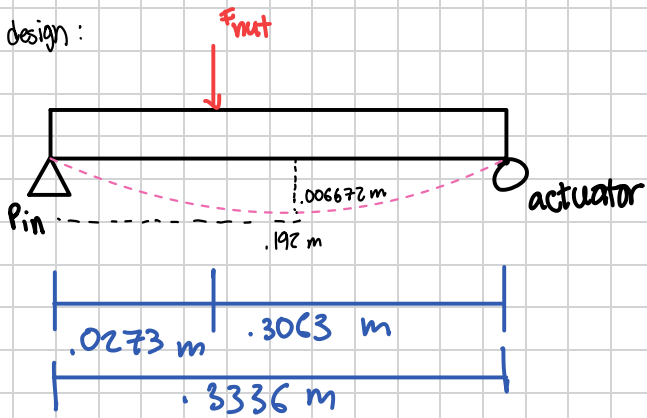
so closest I_x on chart = $.687\times10^6\,mm^4$ for a C channel w/ a cross section of C 75x8.9 mm

↑ in textbook

Beam design: polycarbonate C channel w/ a C75x8.9 mm

Reflection: Finding a beam material that has an acceptable I value was very difficult. Therefore, this material and design for the nutcracker may be unrealistic.

c) Final design:



cross-section: